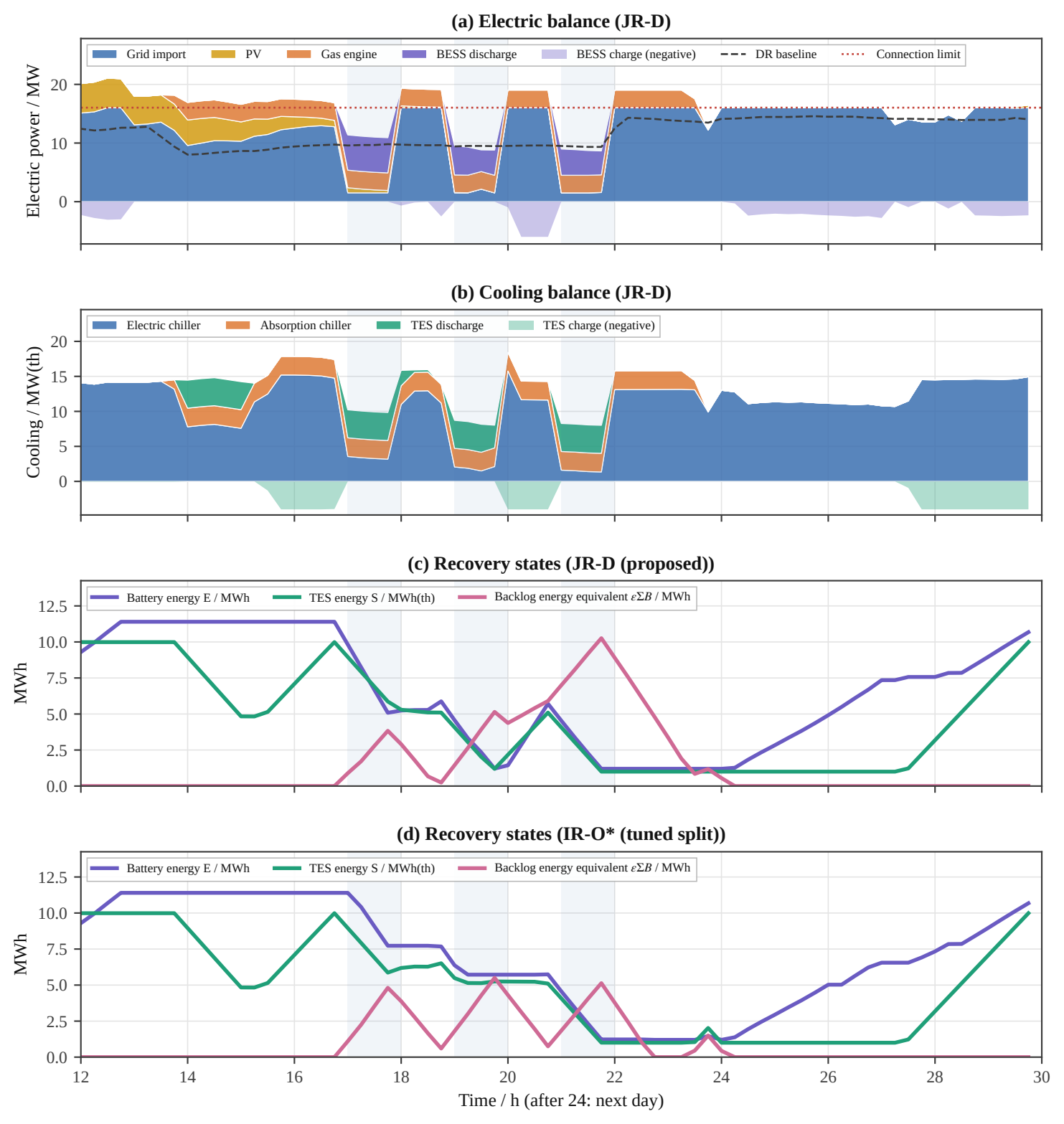
(a) Electric balance (JR-D)
Grid import
PV
Gas engine
BESS discharge
BESS charge (negative)
DR baseline
Connection limit
Electric power / MW
20
10
0
(b) Cooling balance (JR-D)
Electric chiller
Absorption chiller
TES discharge
TES charge (negative)
Cooling / MW(th)
20
15
10
5
0
(c) Recovery states (JR-D (proposed))
Battery energy E / MWh
TES energy S / MWh(th)
Backlog energy equivalent $\epsilon\Sigma B$ / MWh
MWh
12.5
10.0
7.5
5.0
2.5
0.0
(d) Recovery states (IR-O* (tuned split))
Battery energy E / MWh
TES energy S / MWh(th)
Backlog energy equivalent $\epsilon\Sigma B$ / MWh
MWh
12.5
10.0
7.5
5.0
2.5
0.0
12
14
16
18
20
22
24
26
28
30
Time / h (after 24: next day)